QUASI-FINITE SEPARATED MORPHISMS CARRYING A COHERENT MODULE WHOSE PUSHFORWARD IS COHERENT AND ACYCLIC ARE FINITE

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file functors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for functors.tex (fu-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original reference 05K0: sources/stacks-chow/tags.txt, source lines 7184–7184. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0BK0: sources/stacks-chow/tags.txt, source lines 14543–14543. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0AVZ: sources/stacks-chow/tags.txt, source lines 13702–13702. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : V \to X$ be a quasi-finite separated morphism of Noetherian schemes. If there exists a coherent $\mathcal{O}_V$-module $\mathcal{K}$ whose support is V such that $f_*\mathcal{K}$ is coherent and $R^q f_*\mathcal{K} = 0$, then f is finite.*

Proof. By Zariski's main theorem we can find an open immersion $j : V \to Y$ over X with $\pi : Y \to X$ finite, see More on Morphisms, Lemma 05K0. Since π is affine the functor π_* is exact and faithful on the category of coherent $\mathcal{O}_X$-modules. Hence we see that $j_*\mathcal{K}$ is coherent and that $R^q j_*\mathcal{K}$ is zero for $q > 0$. In other words, we reduce to the case discussed in the next paragraph.

Assume f is an open immersion. We may replace X by the scheme theoretic closure of V. Assume $X \setminus V$ is nonempty to get a contradiction. Choose a generic point $\xi \in X \setminus V$ of an irreducible component of $X \setminus V$. Looking at the situation after base change by $\mathrm{Spec}(\mathcal{O}_{X,\xi}) \to X$ using flat base change and using Local Cohomology, Lemma 0BK0 we reduce to the algebra problem discussed in the next paragraph.

Let $(A, \mathfrak{m})$ be a Noetherian local ring. Let M be a finite A-module whose support is $\mathrm{Spec}(A)$. Then $H^i_{\mathfrak{m}}(M) \neq 0$ for some i. This is true by Dualizing Complexes, Lemma 0AVZ and the fact that M is not zero hence has finite depth. □

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